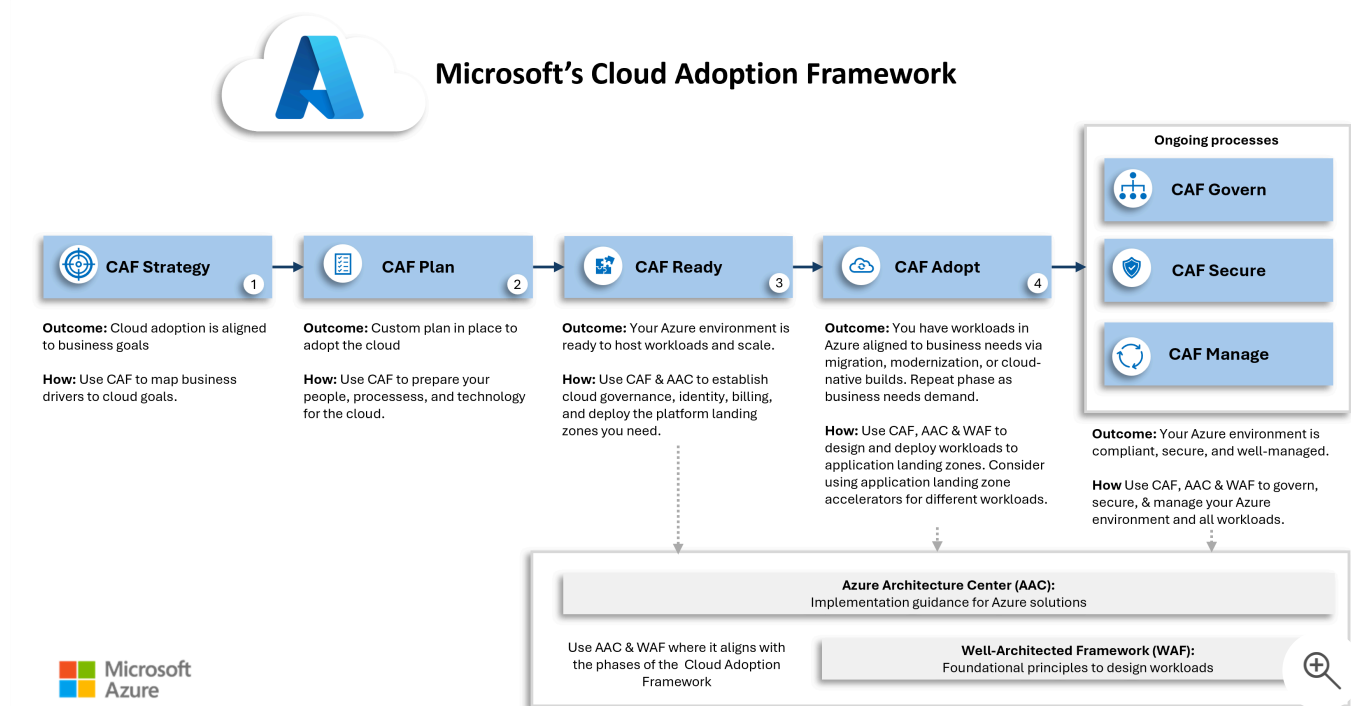


What is the Microsoft Cloud Adoption Framework?

The Microsoft Cloud Adoption Framework (CAF) is a structured roadmap that helps organizations successfully adopt Azure and integrate it into their existing IT environments. It provides best practices and guidance to help organizations make the right decisions for their business throughout the cloud journey.



How does the Cloud Adoption Framework work?

The Cloud Adoption Framework organizes Azure guidance into seven core methodologies that together guide an organization's cloud journey. The foundational methodologies (Strategy, Plan, Ready, and Adopt) are sequential. They help you define business outcomes, prepare your organization and environment, and deploy workloads to Azure. These phases are where most organizations start and are revisited as cloud adoption matures. Once workloads are running in Azure, the operational methodologies (Govern, Secure, and Manage) define different aspects of your cloud operations that help ensure your Azure environment remains compliant, protected, and optimized over time.

CAF methodology (phases)	Outcome	How
 1. Strategy	Cloud adoption aligned to business goals	Map business drivers to cloud outcomes.
 2. Plan	A plan for cloud adoption	Operating model, cloud skills, migration plan, and cloud cost estimation.
 3. Ready	An Azure environment ready for workloads	Azure purchasing, tenant setup, platform landing zone, application landing zones.
 4. Adopt	Workloads in Azure meeting business needs	Migrate, modernize, or build-cloud native workloads.
 5. Govern	Control workloads	Assess cloud risks and mitigate with Azure and Microsoft tools.
 6. Secure	Protect workloads	Apply security control with with Azure and Microsoft tools.
 7. Manage	Optimize workloads	Administer workloads with Azure and Microsoft tools.

Why use the Cloud Adoption Framework?

The Cloud Adoption Framework is about adopting cloud technologies and integrating them into your business. It provides proven practices to govern, secure, and manage all workloads. It provides structured processes for migrating, modernizing, and building cloud-native solutions. Throughout each stage of adoption, the framework guides you to the right tools, architectures, and other resources to help you achieve your organizational goals. The framework integrates with the [Well-Architected Framework](#) and [Azure Architecture Center](#).

Who is the Cloud Adoption Framework for?

The Microsoft Cloud Adoption Framework serves organizations across all industries and scales pursuing strategic Azure cloud adoption. This framework adapts to your specific cloud maturity level, from initial Azure evaluation through advanced multicloud optimization strategies. It's designed for anyone in the organization responsible for technology adoption, whether that responsibility spans business strategy, technology governance, cloud integration, or day-to-day operations.

Organization Type	How the framework accelerates success
Startups	Build cloud-native applications using best practices. Access Microsoft for Startups for Azure credits and technical resources.
Enterprise	Establish a governance foundation to control workloads at enterprise scale.

Cloud Adoption Framework scenarios

The Cloud Adoption Framework has specific guidance aligned to common scenarios across our customer base.

 Expand table

CAF Scenario	Purpose
Data	How to prepare your data for AI and analytics
AI adoption	How to adopt AI (generative and nongenerative) workloads
AI agents	How and when to adopt AI agents
Azure Virtual Desktop	How to adopt Azure Virtual Desktop workloads
Azure VMware Solution	How to adopt Azure VMware Solution workloads
Hybrid and multicloud	How to adopt Azure for hybrid and multicloud
Oracle on Azure	How to adopt Oracle workloads on Azure.
SAP on Azure	How to adopt SAP workloads on Azure

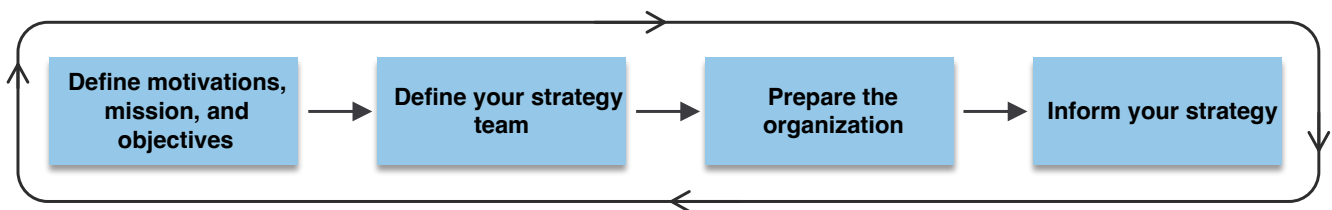
Last updated on 09/05/2026

Develop a cloud adoption strategy

A cloud adoption strategy directs how your business adopts cloud technology. It connects executive intent to measurable outcomes and sets the standards that every workload team operates within.

A sound strategy starts with a clear understanding of your business objectives. From there, it defines the priorities, guardrails, and investment decisions that align cloud initiatives to those objectives.

The Cloud Adoption Framework Strategy methodology guides you through this work in four steps.



Strategy isn't a one-time exercise. Organizations mature through iteration, so treat this guidance as a recurring input to your strategy rather than a one-pass document. Revisit it as business conditions, regulatory requirements, and your cloud footprint evolve.

Whether you're establishing a cloud presence or expanding an existing one, this guidance helps you set a clear mission, define measurable goals, and make informed investment decisions across your portfolio.

Why build a cloud adoption strategy?

A cloud adoption strategy helps you:

- **Maximize the value of cloud investments.** Direct spend toward initiatives that deliver financial efficiency and business agility, and reduce waste from uncoordinated adoption.
- **Align technology decisions to business outcomes.** Tie every cloud initiative to a measurable business objective so investment trade-offs are explicit and defensible.
- **Unify cross-functional leadership.** Establish shared goals across business, IT, finance, and security so teams move in the same direction under a common operating model.

- **Manage portfolio-level risk.** Set a deliberate path for modernizing legacy estates, meeting compliance obligations, and avoiding the security and operational risks of ad-hoc adoption.
- **Sustain improvement over time.** Build a strategy that adapts as conditions change, so the organization continues to address emerging issues such as cost overruns, siloed operations, and unrealized value.

How to build a cloud adoption strategy

This guidance is organized into four steps that apply to enterprises and growth-stage organizations alike:

1. **Define motivations, mission, and objectives.** Establish the business reasons for adopting cloud and the outcomes by which success is measured.
2. **Define your strategy team.** Identify the leaders and stakeholders accountable for setting direction and ensuring execution across the organization.
3. **Prepare the organization.** Secure executive sponsorship, align organizational strategies, evaluate your operating model's fit for cloud, and decide whether to shift toward a product-oriented model.
4. **Inform your strategy.** Apply strategic considerations across the dimensions that shape cloud outcomes: cost efficiency, resiliency, security, and sustainability.

Cloud adoption strategy checklist

The following checklist provides an overview of the tasks for each cloud adoption strategy step. Focus on the startup tasks if your organization is smaller and has fewer resources. Focus on the enterprise tasks if your organization is larger and has extensive resources. Some tasks correspond to both scenarios.

 Expand table

Cloud adoption strategy step	Startup tasks	Enterprise tasks
Define your motivations, mission, and objectives	☐ Define your motivations ☐ Define your mission and objectives ☐ Measure success	☐ Define your motivations ☐ Classify your motivations ☐ Define your mission and objectives ☐ Measure success
Define your strategy team	☐ Define your team	☐ Define your team

Cloud adoption strategy step	Startup tasks	Enterprise tasks
Prepare your organization	☐ Ensure leadership and executive buy-in	☐ Ensure leadership and executive buy-in ☐ Align organizational strategies ☐ Understand your operating model's readiness ☐ Shift from a project model to a product model ☐ Identify and define partner relationships
Inform your strategy	☐ Address security	☐ Ensure cost efficiency ☐ Create resiliency ☐ Address security ☐ Adopt sustainability

Next step

Determine your motivations to move to or expand in the cloud

Last updated on 05/20/2026

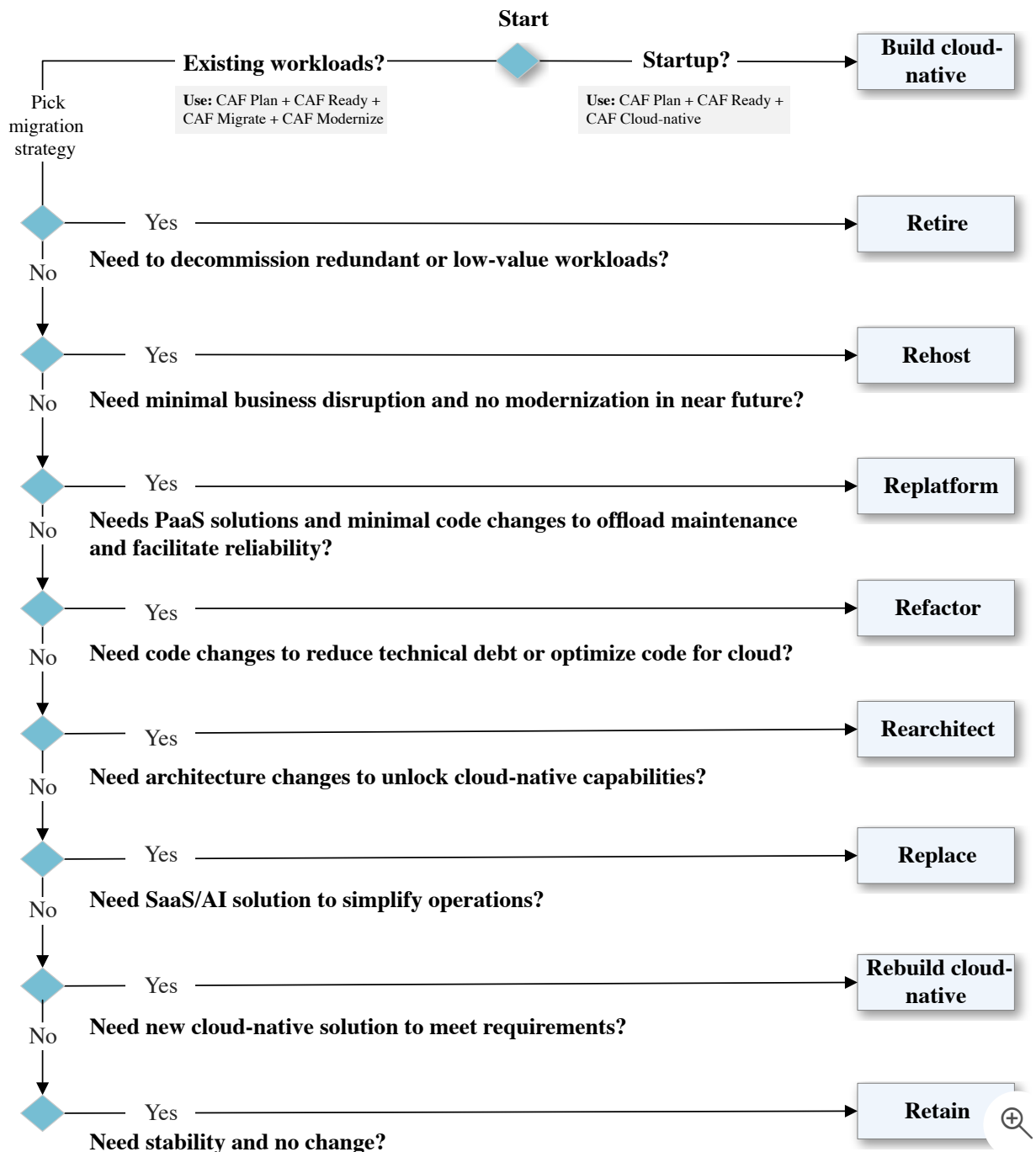
Prepare your organization for the cloud

Successful cloud adoption, whether startup or larger organization, requires more than technical readiness. It requires a cloud adoption plan. A cloud adoption plan converts your cloud strategy into an actionable plan that's specific to your goals. Use this guidance to define your cloud adoption plan.

Map your cloud adoption journey

Every organization's cloud journey is different. Startups often begin cloud-native, while enterprises must assess and migrate existing workloads. Knowing your starting point helps you navigate the Cloud Adoption Framework (CAF) effectively.

1. **Build cloud-native.** If you're a startup with no existing workloads, you build your business cloud-native from the ground up. This approach fosters rapid development using cloud-native design. You follow a portion of CAF Plan: prepare, plan, estimate, document. Then you use [CAF Ready](#) to quickly establish your Azure environment and [CAF Cloud-native](#) to create cloud-native apps.
2. **Migrate existing workloads.** Enterprises must evaluate their existing IT estate and determine the best migration strategy for each workload based on business drivers. You follow all of CAF Plan: prepare, plan, discovery, select, assess, estimate, document. Then, you prepare your enterprise environment using an Azure landing zone in [CAF Ready](#). Then, you execute migration using [CAF Migrate](#) and modernize using [CAF Modernize](#).



Choose a cloud operating model

A cloud operating model defines how your organization manages cloud resources, team responsibilities, and collaboration. Choose a model that fits your current state but also supports future growth. A well-designed operating model helps align cloud efforts with business goals, accelerate workload delivery, clarify accountability, and reduce operational overhead.

1. **"Centralized" cloud operating model:** In a centralized model, a single team is responsible for governance, security, and operations across the entire cloud estate and all workloads.

This model ensures consistent policy enforcement and is ideal for small organizations, startups, or highly regulated industries. However, it can become a bottleneck as cloud adoption scales. Reevaluate and adjust responsibilities regularly to maintain agility.

2. **"Shared management" cloud operating model:** In a shared management model, cloud responsibilities are divided between platform teams and workload teams. Platform teams deliver a standardized set of platform "products" used across the organization. They enforce baseline governance policies and maintain platform landing zones for shared services like connectivity, identity, management, and security. Platform teams also offer platform services as products, such as [subscription vending](#) for application landing zones, developer environments, AI tooling, and data services. Workload teams operate autonomously within these guardrails. This model suits mid-size and enterprise organizations and is especially effective in hybrid or multicloud environments. See [Shared management operations](#).
3. **"Decentralized" cloud operating model:** In a decentralized model, each team owns its platform landing zone and manages its workloads independently, including governance and operations. This approach suits highly skilled teams in startups or innovation programs. To mitigate security and compliance risks, assess team capabilities, provide training, and conduct regular audits.
4. **Understand hybrid cloud operating models.** Many organizations benefit from a hybrid approach, where different teams operate under different models based on their maturity and needs. For example, core systems might follow a centralized or shared model, while innovation teams operate more independently. Evaluate team capabilities and business priorities to determine where hybrid models add value without compromising governance.
5. **Evolve operations as needed.** As your organization scales, reassess whether your current operations approach supports your goals. Regularly reassess whether your current model supports your strategic goals and operational needs.
6. **Engage business and IT leadership.** Successful cloud operations require alignment across the organization. Engage both business and IT leaders to ensure the operations approach reflects shared priorities. This collaboration fosters buy-in, secures necessary resources, and ensures the model has the authority to drive change.

Operations approach	Best for	Responsibilities	Pros	Cons
Centralized	Startups, single-cloud estates	One cloud team manages every function	Simpler control, uniform standards	One team becomes a bottleneck as scale grows
Shared Management	Mid-size or enterprise estates, hybrid, or multicloud	Platform teams build the platform landing zones , manages shared services, and governance guardrails. Workload teams work autonomously within guardrails	Balances standards with team agility	Requires clear responsibilities and strong coordination
Decentralized	Tech-savvy startups, innovation programs	Each team owns every cloud task	High speed and autonomy	Weaker standardization, higher risk of security gaps

Plan your cloud responsibilities

Clearly defined responsibilities are essential for consistent governance, security, and operations across your Azure environment. Assign ownership early to maintain control and accountability.

1. **Plan for cloud governance.** Establish a governance team to assess risks, define policies, and monitor progress. This team should be in place before deploying workloads. See [Build a cloud governance team](#).
2. **Plan for cloud security.** Embed security into every stage of the cloud lifecycle from development to operations. A secure foundation protects against threats and ensures compliance. See [Security Teams, Roles, and Functions](#) to start your security plan.
3. **Plan for cloud management.** Define operational processes to align cloud operations with business goals and ensure consistent service delivery. See [Ready your Azure cloud operations](#) for a detailed list of responsibilities.
4. **Plan for AI adoption.** AI is a transformational technology that every organization needs to prepare for. Start by developing your [AI Strategy](#) and [building an AI team](#). Also review how to [govern AI](#), [secure AI](#), and [manage AI](#).

Document cloud responsibilities

Well-documented cloud responsibilities are essential for clarity, accountability, and operational consistency. It defines who owns what across governance, security, and operations, internally and with partners.

- 1. Map responsibilities across governance, security, and operations.** Assign clear ownership for each function to eliminate ambiguity and ensure comprehensive coverage. Include both primary and backup owners to maintain continuity during absences or transitions. This mapping helps teams understand their roles and reduces the risk of missed tasks or duplicated efforts.
- 2. Define partner roles.** Clearly outline the scope of responsibilities for strategic [partners](#) . Specify which functions partners own, such as platform operations, workload management, or innovation initiatives. Ensure these roles complement internal team responsibilities without overlap.
- 3. Communicate the responsibilities to all stakeholders.** Ensure that every team understands its role and how it interacts with others. Provide training, documentation, and onboarding materials to reinforce clarity and support collaboration. This transparency helps teams work effectively within the operations approach and reduces friction across functions.
- 4. Review responsibilities regularly.** As your Azure environment grows and team capabilities evolve, revisit your responsibilities assignments to ensure they remain aligned with business goals and operational needs.

Azure tools and resources

 Expand table

Category	Tool	Description
Partner solutions	Microsoft Partner Center	Helps identify and connect with trusted partners and expert partners for specialized cloud expertise.
Operations approach	Shared management operations approach	Provides detailed guidance on implementing the shared responsibility model for enterprise organizations.
Governance	Build a cloud governance team	Offers guidance on forming a team to assess risks, document policies, and enforce governance standards.
Security	Security teams, roles, and functions	Details roles and responsibilities for integrating security into the cloud lifecycle.
Manage	Ready your Azure cloud operations	Provides a checklist for preparing operational processes to manage Azure environments effectively.

Next steps

Prepare your people for Azure

Last updated on 09/15/2025

What is an Azure landing zone?

The Ready phase of the Cloud Adoption Framework helps you design your Azure landing zone to meet your organization's requirements. An Azure landing zone is a proven and flexible architecture for governing, securing, and scaling a multi-subscription Azure environment. An Azure landing zone consists of two components:

- **Platform landing zone:** The centralized foundation that establishes governance, security, and shared resources for all your Azure workloads.
- **Application (workload) landing zones:** Azure environments where you deploy and operate workload resources within the standards established by the platform landing zone.

You can implement each component of your Azure landing zone in one of two ways:

- **Accelerators:** Microsoft-provided accelerators for both platform landing zone and application landing zone scenarios use infrastructure as code to deploy Azure environments based on recommended practices.
- **Custom build:** You design and build the solution internally or with support from Microsoft or Microsoft partners.

For most organizations, accelerators provide the fastest path to a deployment that aligns with Microsoft's recommended practices.

Azure landing zone architecture

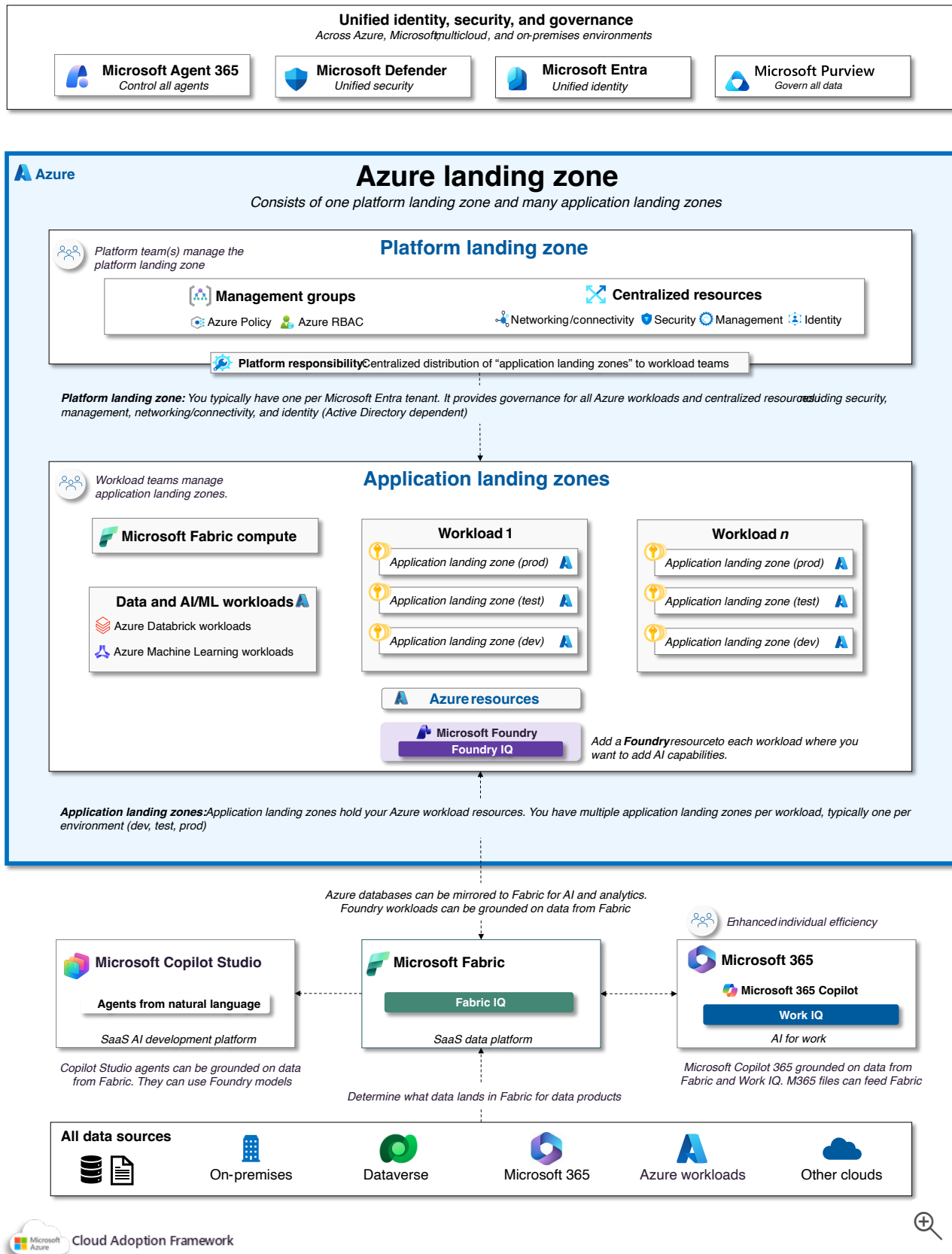
Every Azure landing zone starts with the platform landing zone. Before you build a platform landing zone, define organizational requirements across the Azure landing zone [design areas](#). These decisions determine how you design your Azure landing zone.

The Azure landing zone reference architecture represents a recommended target architecture. Use it as a starting point and adapt it to meet your organization's requirements.

Conceptual

Microsoft's Cloud Adoption Framework

Azure landing zone: Conceptual architecture



Platform landing zone

The platform landing zone establishes your organization's Azure foundation. It defines how you organize your Azure resources, enforce governance, and provide shared capabilities centrally. Most organizations should have only one platform landing zone per Microsoft Entra tenant.

A platform landing zone consists of a management group hierarchy and centralized resources (as needed). An important function of the platform is to have a way to distribute application landing zones to workload teams.

- **Management group hierarchy:** The management group hierarchy provides the governance structure for the Azure environment. It organizes all of your Azure subscriptions and applies governance standards to those subscriptions. The hierarchy separates platform resources and workload resources. A well-designed hierarchy allows you to apply governance standards consistently without creating unnecessary administrative complexity.
- **Centralized resources:** Many organizations provide certain capabilities centrally so every workload can use the same approved services and standards. Common examples include connectivity and networking, security monitoring, Active Directory identity services, and management services. What you centralize depends on your requirements. Only centralize capabilities that provide clear governance, operational, or economic benefits across multiple workloads.
- **Key platform responsibility:** You need a repeatable process for requesting, creating, and distributing application landing zones to workload teams. A standardized approach ensures every new application landing zone follows your organization's governance and security baselines. You can do this process manually or automate it. As the number of application landing zone requests grows, automation becomes increasingly important. See [Subscription vending](#).

Implementation options: Microsoft provides several platform landing zone accelerators that implement recommended architecture patterns and organizational guardrails. See [Platform landing zone implementation options](#). You can also develop your own implementation or work with Microsoft or Microsoft partners to build a customized platform landing zone.

Application landing zones

Application landing zones host Azure workload resources. Most workloads use multiple application landing zones, with separate application landing zones for each environment such as development, testing, and production. An application landing zone can be one or more Azure

subscriptions. You only need multiple Azure subscriptions when required to accommodate workload requirements, organizational boundaries, or Azure subscription limits.

The platform team places application landing zones in either the Online, Internal ("Corp"), or Local management group under the (application) "Landing zones" management group. The application landing zones inherit the governance and security policies from the Azure Policies applied to the management group hierarchy. This model provides consistent guardrails and allows flexibility to meet workload-specific requirements.

Implementation options: Microsoft provides several accelerators to establish application landing zones. You can also implement their own approach or engage Microsoft or Microsoft partners for assistance. See the [Azure Architecture Center](#) for a list of application landing zone accelerators.

Technology adoption in an Azure landing zone

An Azure landing zone supports new and emerging technologies without requiring changes to the overall architecture. You deploy workload resources into application landing zones regardless of workload type, such as AI. As new requirements emerge, you update the governance and security policies in the platform landing zone. Those policies are then applied consistently across application landing zones, which helps you adopt new technologies faster and maintain consistency across all workloads of that type.

The Cloud Adoption Framework provides guidance for common adoption scenarios to help you prepare your people, processes, and platform for successful adoption. See [Scenarios](#).

Next steps

[Design principles](#)

Last updated on 07/29/2026

Build a cloud governance team

Cloud governance is how an organization controls its use of cloud services by establishing guardrails. These guardrails are a set of policies, procedures, and tools that define acceptable and unacceptable cloud activities. Effective cloud governance aligns cloud usage with business objectives, mitigates risks, ensures regulatory compliance, and prevents unmanaged or unauthorized cloud actions. In practice, cloud governance covers key domains, such as security, regulatory compliance, operations, cost management, data management, resource provisioning, and even emerging areas like AI.

Quick links: [Azure governance tools](#) and [Example governance RACI matrix](#)

Governance process: Cloud governance is not a one-time project, but a continuous process. After initial setup, it requires ongoing monitoring, evaluation, and updates to adapt to new technologies, evolving risks, and changing requirements. After establishing the governance foundation in Step 1, Steps 2-5 repeat in a cycle to sustain and improve governance over time.



The first step is to establish a dedicated cloud governance team to oversee cloud governance across the organization. This team is accountable for managing cloud-related risks, developing and updating governance policies, and reporting on governance progress. Team members must understand the needs of various business units and ensure that governance policies effectively minimize risk without blocking business goals. The outcome of this step is to have clear ownership and accountability for cloud governance success.

1. Define the team's function

Clearly outline what the cloud governance team is responsible for and what activities it will perform to implement governance. At a minimum, the team should be able to:

1. **Engage stakeholders.** The cloud governance team must actively engage stakeholders across the organization (IT, finance, operations, security, and compliance) to gather input on defining cloud governance policies. The goal is to ensure cloud governance policies minimize risk without preventing teams from achieving business goals.

2. **Assess cloud risks.** The cloud governance team must oversee the identification and evaluation of cloud-related risks. The team drives risk assessment processes and communicates risk findings. They might provide tools or frameworks for others to assess security, compliance, and operational risks in the cloud.
3. **Develop and update governance policies.** The cloud governance team should document the cloud governance policies that address identified risks. The team resolves any challenges these policies create for various groups and periodically reviews policies to keep them up to date with technological changes and new requirements. The policies must be comprehensive, enforceable, and aligned with current business needs.
4. **Monitor and review compliance.** The cloud governance team needs to establish metrics and reporting methods to measure how effective the governance policies are. Track compliance levels, policy violations, incident response times, and even user satisfaction. Regularly review these metrics to identify areas for improvement and report on the organization's cloud governance posture.

2. Select team members

Choose individuals for the cloud governance team who have the right skills and experience to enforce policies, manage risks, and ensure compliance. Some recommendations for team composition:

1. **Keep a small team.** Pick a small team to encourage agility and quicker decision-making.
2. **Ensure diverse representation.** Include team members from different departments or domains. For example, IT operations, cloud architecture, security, compliance, finance, and perhaps application development. Cross-functional representation ensures that the governance policies consider multiple perspectives.
3. **Define team members' responsibilities.** Define the roles and responsibilities within your cloud governance team. Tailor them to your organization's size, complexity, and cloud maturity. Key responsibility areas typically include overall cloud governance program success, cloud architecture oversight, cloud security, regulatory compliance, and cloud financial management (cost optimization).

3. Define the team's authority

Empower the cloud governance team with the mandate and support required to implement governance throughout the organization. Steps to do this include:

- 1. **Secure executive sponsorship.** Gain support from and report to a named executive, such as the CIO or CTO, to support the cloud governance initiative. The executive sponsor serves as a point of escalation for challenges and helps align cloud governance with business goals.
- 2. **Establish authority levels.** The executive sponsor should grant the team the authority to define cloud governance policies and take corrective measures for noncompliance.
- 3. **Communicate authority.** The executive sponsor should communicate the authority of the cloud governance team to the entire organization. Include the importance of adhering to the cloud governance policies they create.

4. Define the team's scope

Establish the boundaries of the cloud governance team's responsibilities. The goal is to clarify areas of responsibility so the cloud governance team can focus on their defined functions. To define the scope, follow these recommendations:

- 1. **Define relationship with other teams.** Determine how the cloud governance team will interact with existing IT governance, on-premises infrastructure teams, or application teams. For example, in a hybrid environment, specify which aspects the cloud governance team handles versus traditional IT governance. Clearly delineating scope prevents confusion over who manages what.
- 2. **Use a RACI matrix.** It can be helpful to create a RACI (Responsible, Accountable, Consulted, Informed) chart that maps out governance-related tasks and who is involved. For instance, the cloud governance team might be Accountable for developing policies, while cloud platform engineers are Responsible for implementing specific controls, and so on. A RACI matrix ensures everyone knows their part in cloud governance and how the governance team collaborates with other groups.

Example cloud governance RACI matrix

The following table is an example of a RACI matrix for cloud governance. The matrix indicates who is responsible (R), accountable (A), consulted (C), and informed (I) across various cloud governance tasks. Create a RACI matrix that aligns to your organization and meets your specific needs.

Task	Cloud governance team	Executive sponsor	Cloud platform team(s)	Workload teams
Engage stakeholders	R, A	I	C	C
Assess cloud risks	A	I	R	R
Develop and update governance policies	R, A	I	C	C
Report on cloud governance progress	R, A	I	C	C
Plan a cloud architecture	A	I	R	R
Enforce governance policies	A, C	I	R	R
Monitor compliance	A, C	I	R	R

Having such a matrix helps clarify who does what. For example, the Cloud Governance Team is accountable (A) for assessing cloud risks, but the Cloud Platform and Workload teams are responsible (R) for performing risk assessments in their areas. The Cloud Governance team is consulted (C) or accountable for enforcement, but the platform and workload teams carry it out.

With the team's functions, membership, authority, and scope established, you have created a foundation for cloud governance. This team will now drive the next steps: defining policies, enforcing them, and monitoring compliance.

Next step

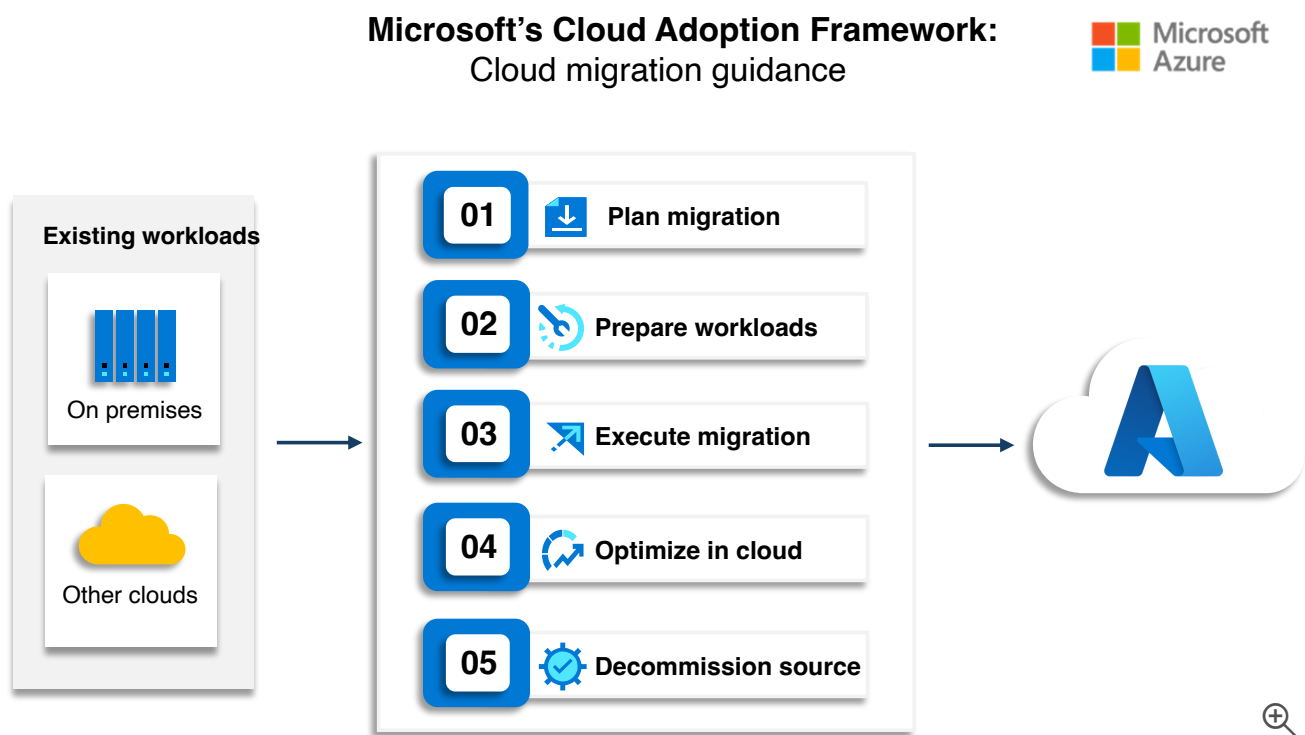
Assess cloud risks

Last updated on 09/18/2025

Plan your migration

A migration plan defines the specific order, timing, and approach for migrating workloads to Azure. This plan translates high-level migration strategies into actionable deployment sequences. It builds on the cloud adoption plan by addressing tactical decisions such as workload prioritization, migration sequencing, and data transfer methods.

Prerequisites: [Migration adoption plan](#), [Azure landing zone](#)



Assess migration readiness and skills

A readiness assessment ensures that your team has the skills and support needed to execute the migration plan. This step identifies capability gaps and accelerates progress through targeted training or external support.

1. **Evaluate your team's Azure skills.** Review your team's experience with Azure services, migration tools, and cutover. This evaluation helps you identify specific knowledge gaps and determine what training your team needs to succeed.
2. **Engage external expertise when needed.** If your team lacks experience with cloud migration, bring in [Microsoft or a Microsoft partner](#) . External experts can validate your

migration strategy, recommend appropriate tools, and help establish realistic timelines. This support reduces risk and speeds up your migration, especially for complex or large-scale projects.

Choose your data migration path

A data migration path is how you move data from your current location to Azure. The right path ensures your data transfers securely, quickly, and cost-effectively. First, check what network connections you have available, ExpressRoute, VPN, or public internet, to understand your options.

- 1. **Use ExpressRoute when you have it.** ExpressRoute gives you a private, dedicated connection to Azure that's faster and more secure than internet connections. If you already have ExpressRoute or plan to get it, use this method for all your workloads. Keep in mind that ExpressRoute requires setup time and has data transfer costs.
- 2. **Use VPN if ExpressRoute isn't available.** Choose a VPN when you need secure data transfer but don't have ExpressRoute. A VPN creates an encrypted tunnel over the internet to Azure, though it's typically slower than ExpressRoute. Make sure you have a VPN Gateway configured in Azure before starting.
- 3. **Use Azure Data Box for large amounts of data.** Data Box is best for offline migrations with lots of data. Microsoft ships you a physical device to copy your data onto, then you ship it back. This option avoids using your network but takes the longest due to shipping time.
- 4. **Use public internet for less sensitive data.** This option works when your data doesn't need encryption and you can't use ExpressRoute or Data Box. While this method is available everywhere, it's the least secure and can slow down your other internet activities.

 Expand table

Data Migration Path	When to use	Pros	Cons
ExpressRoute	Any workload when available	Secure and fast	Setup required, costs money
VPN	Secure transfers when no ExpressRoute	More secure than public internet	Requires setup, slower than ExpressRoute
Azure Data Box	Offline migration with large amounts of data	Moves data without using your network	Slowest method due to shipping

Data Migration Path	When to use	Pros	Cons
Public internet	Nonsensitive data and can't use Data Box	Works everywhere	Least secure, uses your bandwidth

Determine the migration sequence

Migration sequencing reduces risk and builds team confidence by establishing a logical order for workload migration. The sequence determines which workloads move first and how dependent components migrate together to prevent service disruptions. Organize large portfolios into migration waves. For detailed guidance on wave planning, see [Migration wave planning](#).

Find dependencies

- 1. Discover all dependencies first.** Dependencies between workloads cause service disruptions if not migrated together. [Map internal and external dependencies](#) to discover these connections before creating migration groups.
- 2. Analyze dependency types and criticality.** Different dependency types require different migration approaches. Distinguish between these categories:

 Expand table

Dependency type	Description	Migration approach
Direct dependencies	Require immediate communication and low latency between components.	Move all directly connected components together to maintain performance and avoid disruptions.
Indirect dependencies	Involve occasional or non-critical interactions between systems.	Migrate together or in separate waves if the connection tolerates latency or supports hybrid use.
Business dependencies	Depend on organizational or management relationships.	Group and migrate related workloads and reporting systems together to align with business priorities.

- 3. Group workloads by dependency relationships.** Create groups based on shared databases, APIs, authentication services, or network connections. These groups form the foundation of your migration waves and ensure all components required for functionality move together.

When uncertainty exists about dependency criticality, group the components together. This conservative approach provides flexibility for future separation.

4. **Document each dependency group systematically.** Tag assets based on their dependency groups using consistent naming conventions. Document each group with:
 - **Group name and ID** - Unique identifier and descriptive name
 - **Component inventory** - All infrastructure elements, applications, and services
 - **Critical dependencies** - Essential connections requiring special handling
 - **Migration constraints** - Business, technical, or timing requirements
5. **Validate group completeness.** Confirm each group includes all necessary components for applications to operate, including supporting infrastructure such as load balancers, DNS records, or caching layers.

Address split-environment operations

1. **Plan for unmovable dependencies.** Identify components that must stay in your source environment due to technical or regulatory reasons. Document why they can't move, how they connect to other systems, and what data they share. This documentation helps you create strategies for these components to work smoothly with your cloud systems.
2. **Minimize split-environment operation time.** When components can move to the cloud later but not immediately, document their connections and data flows with cloud systems. Create a clear plan with timelines and risk management approaches to reduce the time your workload operates across both environments. Consider delaying the migration until more components can move together.
3. **Connect environments effectively.** Use integration methods like API gateways, message queues, and data synchronization to create reliable connections between your cloud workloads and source environment components. These approaches reduce delays, improve security, and prepare the way for eventually moving remaining components to the cloud.

Prioritize workloads to migrate

1. **Review workload details.** Work with stakeholders to review business and technical details for each workload. Ensure that downtime or failure impacts are well understood and align with current business priorities. Use the [migration adoption plan](#) to verify details like business unit, workload owner, technical dependencies, and criticality classification. These details help prioritize and sequence workloads effectively.

Priority	Business value	Effort	Description
High	High	Low	Quick wins - migrate first for immediate impact
Medium-High	High	High	Strategic investments - plan carefully with adequate resources
Medium-Low	Low	Low	Easy candidates - fill gaps between major migrations
Low	Low	High	Avoid or defer - focus resources on higher-value opportunities

2. **Start with simpler workloads to reduce risk.** Begin migrating workloads that are less complex and have lower risk. This approach helps your team gain confidence and refine migration processes before tackling more challenging workloads. Target internal tools, development environments, or low-usage applications with standalone architectures and minimal integration points.
3. **Move non-production environments before production.** Nonproduction environments provide a safe space to test the full migration process. Migrate development, staging, and QA environments before production to validate readiness. This order allows teams to test configurations, performance, and recovery procedures without affecting users. Use nonproduction migrations to train operations teams.
4. **Schedule critical systems after you demonstrate initial success.** Critical applications require proven migration capabilities before you move them to Azure. Plan these migrations for later waves when your team demonstrates competency with Azure services. Business deadlines such as hardware refresh cycles might require you to prioritize critical applications earlier with more safeguards and extended testing periods.
5. **Include representative complex workloads to test scenarios.** Add one or two complex workloads to each early wave to expose challenges you face with mission-critical applications. Choose workloads that represent common patterns such as multi-tier applications or database-dependent systems.

Create a detailed migration schedule

1. **Set start and end dates for each migration.** Include buffer time for testing and issue resolution to ensure smooth execution. This detailed scheduling reduces the risk of delays and supports effective resource planning.

- 2. **Align timelines with business events.** Avoid scheduling migrations during critical business periods such as financial close, product launches, or holiday seasons. This alignment reduces the risk of business disruption and ensures stakeholder confidence.
- 3. **Use project management tools to track progress.** Use tools like Azure DevOps to manage dependencies, track milestones, and communicate changes effectively. These tools provide visibility into migration progress and support proactive issue resolution.

Choose the migration method for each workload

Migration methods fall into two categories: migration with downtime and migration with near-zero downtime. Choose the best migration method for each workload based on its downtime tolerance and business criticality.

- 1. **Choose downtime migration for workloads that tolerate planned outages.** Downtime migration is simpler and faster because it doesn't require real-time synchronization between source and target environments. This method works well for noncritical workloads such as development environments, test systems, or applications with scheduled maintenance windows. Document the acceptable downtime duration for each workload and schedule migrations during low-usage periods to minimize business effects.
- 2. **Choose near-zero downtime migration for critical workloads.** Near-zero downtime migration ensures that critical workloads remain operational during the transition through continuous data replication and cutover techniques. This method is essential for customer-facing applications, real-time transaction systems, or workloads with strict service level agreements. Validate that the workload architecture supports continuous replication and that network bandwidth can handle real-time data transfer. Test connectivity and replication processes in a nonproduction environment to confirm readiness for this migration method.

[Expand table](#)

Migration Method	When to use	Pros	Cons
Downtime migration	Noncritical workloads, development environments	Simpler process, faster execution	Service interruption required
Near-zero downtime migration	Critical workloads, strict SLAs	Minimal service disruption	Complex setup and requires testing

Define rollback plan

A rollback plan enables teams to quickly reverse changes when a deployment fails or introduces risk. A well-defined plan minimizes downtime, limits business impact, and maintains system reliability. Always establish rollback criteria and procedures before initiating any migration or deployment.

1. **Define failed deployment.** Collaborate with business stakeholders, workload owners, and operations teams to decide what counts as a failed deployment. Examples include failed health checks, poor performance, security issues, or unmet success metrics. This definition ensures rollback decisions align with your organization's risk tolerance. Include specific conditions that trigger a rollback in your deployment plan, such as CPU usage limits, response time thresholds, or error rates. This evaluation makes rollback decisions clear and consistent during incidents.
2. **Automate rollback steps in CI/CD pipelines.** Use tools like [Azure Pipelines](#) or [GitHub Actions](#) to automate rollback tasks. For example, set up pipelines to redeploy a prior version if health checks fail.
3. **Create workload-specific rollback instructions.** Write rollback steps that match your workload type, environment, and deployment method. For example, IaC deployments require reapplying past templates. App rollbacks involve redeploying a prior container image. Attach rollback scripts, configuration snapshots, and IaC templates to your rollback plan. These assets speed up the process and reduce the need for manual steps.
4. **Test rollback procedures.** Simulate deployment failures in a staging environment to confirm your rollback works. Find and fix gaps in automation, permissions, or dependencies. Verify that the rollback procedure restores the system to a stable, known-good state.
5. **Improve rollback strategies.** After each deployment or rollback, review what worked and what didn't. Update rollback rules, steps, and scripts based on lessons learned. Factor in design changes or new tools. Keep your guides updated so rollback plans stay current and useful.

Engage stakeholders on migration plan

Stakeholder approval validates that your migration plan meets business requirements and risk tolerance. You should secure formal approval before executing migrations.

1. **Document the migration plan with business justification.** Create a structured plan showing workload name, owner, criticality, migration method, downtime window, and business effects. Include rationale for each approach and explain how it minimizes risk.

2. **Present tested rollback procedures.** Show specific rollback plans with steps, timeframes, and success criteria. Include automated and manual capabilities. Document preproduction test results to prove quick service restoration.
3. **Validate schedules against business constraints.** Review timelines with stakeholders to avoid critical business periods, maintenance freezes, and seasonal peaks. Provide alternative options with trade-offs if conflicts exist.
4. **Obtain formal approval and rollback authority.** Secure written approval from stakeholders for the migration plan and rollback procedures. Define decision-making authority and establish emergency communication channels.
5. **Define success criteria and review checkpoints.** Set measurable metrics including performance benchmarks, functionality validation, and user acceptance criteria. Schedule formal review points for go/no-go decisions.

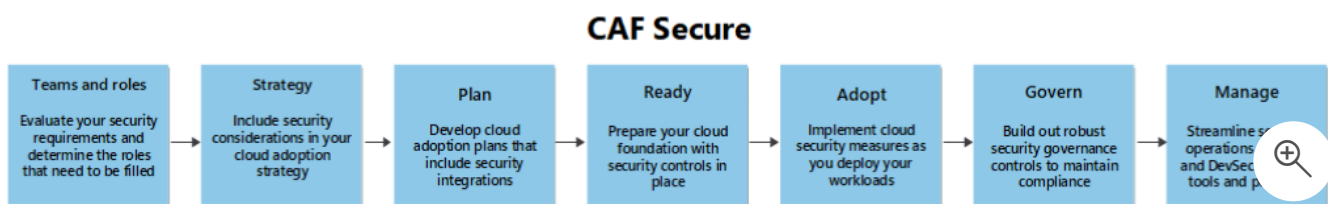
Next step

Prepare workloads for the cloud

Last updated on 08/01/2025

Secure overview

The Cloud Adoption Framework for Azure Secure methodology provides a structured approach for securing your Azure cloud estate. This overview introduces security guidance that applies across every methodology phase of the Cloud Adoption Framework. Security spans strategy, planning, readiness, adoption, governance, and operations; gaps in any phase weaken overall posture. You should apply the Secure methodology end to end so every phase decision reinforces protection, detection, and resilience.



Modernize security posture

Security posture modernization is the continuous elevation of your defenses, detections, and resilience capabilities. This modernization matters here because static controls degrade quickly against evolving attacker tradecraft. You should align modernization work with the [Microsoft Zero Trust adoption framework](#) and enrich each phase task with Zero Trust improvements.

Integrate identity strengthening, segmentation, just-in-time and least-privilege access, threat detection tuning, data protection, and platform baseline automation into your landing zones and operations. Prioritize modernization sprints based on measurable risk reduction (for example, exposed privileges, insecure configurations, unmonitored assets). Automate validation through policy, infrastructure as code, continuous compliance scanning, and secure score tracking in Microsoft Defender for Cloud.

Prepare for and respond to incidents

Incident preparation and response form a primary control layer that limits attacker dwell time and business disruption. This capability matters because even mature preventive controls can't eliminate intrusion attempts. You should implement and continuously improve an end-to-end incident lifecycle covering readiness, detection, triage, containment, eradication, recovery, and post-incident learning.

Codify roles, communication channels, evidence handling, and decision authority. Instrument telemetry ingestion and alert fidelity improvements to cut false positives and accelerate mean time to detect (MTTD). Use Azure incident response guidance to refine runbooks, practice tabletop simulations, and automate containment actions (for example, isolate hosts, revoke tokens, quarantine storage) through orchestrated workflows.

Adopt CIA Triad principles

CIA Triad principles (confidentiality, integrity, availability) provide a concise model for comprehensive information protection. This model matters here because gaps in any single principle create cascading weaknesses. You should map controls, processes, telemetry, and metrics explicitly to each principle for every phase.

- **Confidentiality** restricts access to sensitive data; encryption, key management, identity, access policies, network segmentation, and data classification controls enforce it.
- **Integrity** preserves data correctness and completeness; hashing, signing, immutable storage patterns, version control, and secure update supply chains enforce trustworthy state.
- **Availability** maintains timely access to services and data; redundancy design, fault domain isolation, autoscaling, health probes, chaos testing, backup, and disaster recovery orchestration sustain accessibility.

Apply the triad to drive:

- **Data protection:** Map sensitivity labels and encryption controls to confidentiality risk.
- **Business continuity:** Engineer integrity and availability safeguards to sustain operations.
- **Stakeholder confidence:** Demonstrate measurable adherence to each principle in audits and compliance reporting.

Each article surfaces tasks that address confidentiality, integrity, and availability so you can embed these principles into strategy, design, build, governance, and operations.

Sustain security posture

Security posture sustainment is the disciplined cycle of measuring, improving, and validating control efficacy. This sustainment matters because threat actors iterate rapidly and static defenses lose relevance. You should institutionalize recurring assessment, prioritized remediation, control automation, and evidence-based reporting.

Track [secure score security controls](#) in Microsoft Defender for Cloud to quantify gaps, and couple them with risk-based metrics (for example, exposure of high-privilege identities or unencrypted sensitive stores). Automate drift detection through policy, configuration baselines, and deployment pipelines. Feed incident retrospectives and threat intelligence into backlog refinement so posture change aligns with live adversary behaviors.

Use the cloud security checklist

The cloud security checklist is a consolidated navigation and tracking aid for Secure methodology execution. This checklist matters because it reduces omission risk, accelerates onboarding, and supports audit readiness. You should integrate it into team working agreements and progress reviews as the authoritative task register.

 Expand table

Cloud security step	Cloud security tasks
☐ Understand security teams and roles.	☐ Understand the role of the cloud service provider. ☐ Understand the roles of Infrastructure and Platform teams. ☐ Understand the roles of Security architecture, engineering, posture management teams. ☐ Understand the roles of the Security Operations (SecOps and SOC) teams. ☐ Understand the roles of Security Governance, Risk, and Compliance (GRC) teams. ☐ Learn about security education and policy.
☐ Integrate security into your cloud adoption strategy.	☐ Security posture modernization strategy. ☐ Incident preparedness and response strategy. ☐ Confidentiality strategy. ☐ Integrity strategy. ☐ Availability strategy. ☐ Security posture sustainment strategy
☐ Plan for a secure cloud adoption.	☐ Plan for landing zone adoption. ☐ Security posture modernization planning. ☐ Incident preparedness and response planning. ☐ Confidentiality planning. ☐ Integrity planning ☐ Availability planning ☐ Security posture sustainment planning
☐ Ready your secure cloud estate.	☐ Ready for security posture modernization. ☐ Ready for incident preparedness and response. ☐ Ready for confidentiality. ☐ Ready for integrity.

Cloud security step	Cloud security tasks
	□ Ready for availability □ Ready for security posture sustainment
□ Perform your cloud adoption securely.	□ Security posture modernization adoption. □ Adopt incident preparedness and response. □ Adopt confidentiality. □ Adopt integrity. □ Adopt availability. □ Adopt security posture sustainment
□ Securely govern your cloud estate.	□ Security posture modernization. □ Incident preparedness and response governance □ Confidentiality governance. □ Integrity governance. □ Availability governance. □ Sustaining security governance
□ Securely manage your cloud estate.	□ Security posture modernization. □ Managing incident preparedness and response □ Managing confidentiality. □ Managing integrity. □ Managing availability. □ Managing security sustainment

Next step

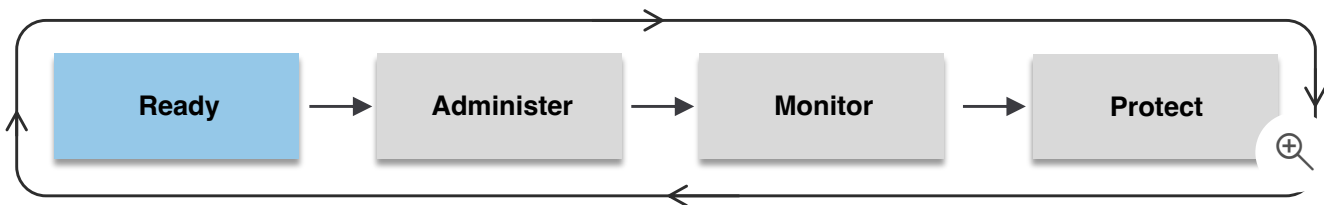
Security teams, roles, and functions

Last updated on 09/15/2025

Ready your Azure cloud operations

Managing your Azure environment isn't just about keeping the lights on. It's about maintaining governance and security while ensuring your cloud stays aligned with business goals over time. As your environment grows, you need to prevent configuration drift, improve deployments through infrastructure as code, and manage change effectively.

The Cloud Adoption Framework (CAF) Manage methodology introduces the RAMP approach (Ready, Administer, Monitor, Protect) to help you build these capabilities. RAMP provides a structured way to organize teams, define responsibilities, and implement the processes and tools that keep your Azure operations secure, compliant, and resilient. From day-to-day administration to monitoring health and protecting against disruptions, RAMP helps you establish a strong operational foundation for long-term success.



Identify your management responsibilities

Effective Azure management spans two layers of accountability: **central (platform)** responsibilities and **workload** responsibilities. Central responsibilities apply across the entire cloud estate. Workload responsibilities focus on individual applications or services. Use *Table 1* to ensure your operating model covers the essential areas of cloud management

Table 1. Primary cloud management responsibilities

 Expand table

Cloud management areas	Central (platform) responsibilities	Workload responsibilities
Compliance	▪ Define operational procedures. ▪ Enforce governance policies. ▪ Monitor compliance and remediate or escalate as required.	▪ Follow operational procedures. ▪ Align design with governance policies.

Cloud management areas	Central (platform) responsibilities	Workload responsibilities
Security	▪ Manage organization-wide security operations. ▪ Manage identities in Microsoft Entra ID. ▪ Grant access to Azure subscriptions. ▪ Define and maintain security baselines via Azure Policy and Microsoft Defender for Cloud. ▪ Oversee threat protection and incident response integration with Microsoft Sentinel.	▪ Implement secure workload design. ▪ Respond to workload-specific security alerts and incidents. ▪ Continuously assess vulnerabilities within the workload.
Resource management	▪ Define and maintain resource hierarchy. ▪ Create workload subscriptions as requested. ▪ Define naming and tagging strategy. ▪ Define network topology. ▪ Configure shared networking (virtual network peering, on-premises connectivity). ▪ Manage cross-workload or shared resources/services. ▪ Monitor subscription limits and handle requests for quota increases.	▪ Manage workload-specific subscriptions (if delegated). ▪ Manage resource groups and resources for each workload. ▪ Adhere to and apply naming and tagging standards. ▪ Manage application-level resource utilization, ensuring resources remain within subscription quotas.
Deployment	▪ Standardize and govern CI/CD pipelines and tools (Azure DevOps, GitHub Actions). ▪ Define reference infrastructure-as-code templates (Bicep, Terraform, ARM templates). ▪ Provide central best practices for pipeline security (code scanning, secrets management).	▪ Use the central CI/CD framework and IaC templates for workload deployments. ▪ Implement workload-specific deployment tasks (configure app settings, database). ▪ Adapt reference templates to workload needs while respecting central guidelines.
Development	▪ Provide and enforce standardized development toolchains and frameworks to accelerate consistency (coding standards, DevOps best practices). ▪ Maintain internal repositories or package feeds for shared libraries or modules.	▪ Adopt and adapt standard toolchains for workload development. ▪ Own the application lifecycle and incorporate best practices (unit testing, integration testing). ▪ Manage continuous improvement for the workload's code base.
Monitoring	▪ Plan monitoring strategy. ▪ Alert on centralized responsibilities. ▪ Provide dashboards for common operational metrics across the environment.	▪ Monitor workload ▪ Extend or fine-tune central alerts to capture workload-specific conditions. ▪ Investigate and remediate workload-level incidents based on alerts and logs.

Cloud management areas	Central (platform) responsibilities	Workload responsibilities
Cost	▪ Allocate global or subscription-level cloud budgets ▪ Monitor organization-wide cloud spend and create cost reports. ▪ Allocate costs to business units or products, typically using tags or custom cost allocation models. ▪ Apply tagging strategy for cost allocation.	▪ Cost optimize workload design ▪ Respect budget constraints.
Reliability	▪ Define reliability requirements (SLO, RPO, RTO) per workload priority. ▪ Provide guidance on business continuity and disaster recovery (BCDR). ▪ Manage centralized disaster recovery solutions. ▪ Support major incident management across all workloads.	▪ Design workload to meet reliability requirements.
Performance	▪ Monitor and maintain performance at centralized components (hub network, shared services). ▪ Provide guidelines for performance optimization and capacity planning. ▪ Monitor quota usage.	▪ Design workload for performance efficiency.

Establish your cloud operations

Use the responsibilities outlined in *Table 1* to build an effective operational foundation. Clearly define teams, standards, and processes by following these steps:

1. **Define your cloud operations model.** Choose a centralized, shared management, or decentralized cloud operating model based on your organization's size and maturity. For guidance, see [Choose a cloud operating model](#)
2. **Assign central platform responsibilities.** Form a dedicated team to handle central management tasks. Develop a skills matrix from *Table 1* to identify required expertise.
3. **Assign workload responsibilities.** Set up specialized teams for workload-specific tasks. Identify responsibilities using *Table 1* then recruit accordingly. Use the Azure Well-Architected Framework and its [Operational excellence](#) pillar to guide your workload management responsibilities.

4. **Assign responsibility.** Name specific owners for all cloud management responsibilities. In a shared management model, workload teams should have autonomy to manage their subscriptions.

Document your cloud operations

Clearly document your cloud operations to enable efficient crisis response and smooth implementation of changes. Establish overarching procedures and create detailed guides for frequent and specific tasks.

Document operational procedures

Define operational procedures for managing change, disaster recovery, and routine maintenance tasks that automation can't handle. Follow these steps:

1. **Define change management procedures.** Change is the major cause of failure in the cloud. Develop a standardized process for managing changes to avoid failures in your cloud environment. See [Manage change](#).
2. **Define deployment procedures (release management).** To maintain consistent configuration, standardize your deployments, releases, and environment promotions. See [Manage deployments](#).
3. **Define disaster recovery and business continuity procedures.** To handle potential failures, prepare a standardized response plan. See [Manage disaster recovery and business continuity](#).
4. **Define additional procedures.** Document processes for managing service requests, patching, and configuration management. Clearly document these processes to ensure stakeholders know how to initiate or complete each task.

Document operational guides

Create detailed step-by-step guides (runbooks or playbooks) for key operational tasks. This preparation ensures consistent execution, improves efficiency, and shortens resolution times during critical events.

1. **Define daily tasks.** Prepare manuals covering daily responsibilities, such as privilege escalation requests and log reviews. Establish standard operating procedures (SOPs) for monitoring metrics, alert thresholds, and dashboards for each system.

2. *Create a library of Azure-centric runbooks.* Create Azure-specific runbooks addressing scenarios such as:

[Expand table](#)

Scenario	Example
High CPU usage	Manage scale up in Azure App Service
Failover and failback	Failover and failback in Azure Site Recovery
Blue/green deployments	Blue/green deployment in Azure Front Door
Backup restoration	Backup restore in Azure Blob Storage and Azure Cosmos DB

3. *Store these runbooks in a central repository.* Maintain runbooks in a central repository accessible by on-call engineers for immediate use during incidents.
4. *Implement operations programmatically.* Integrate infrastructure-as-code into your runbooks to deploy common resources consistently and accurately each time.
5. *Review and update.* Periodically review and revise documentation to reflect operational adjustments and cloud service updates.

Document tools and solutions

Clear documentation ensures consistency, reduces operational risks, and enhances team efficiency. Create and maintain comprehensive documentation for cloud tools. Regularly update documentation to reflect current practices and ensure easy accessibility for all team members.

[Expand table](#)

Area	Example benefits
Integration	Standardization simplifies integrations by consolidating logs and code repositories.
Automation	Reuse IaC templates across teams, automation scripts, and best practices across projects.
Incident management	Capture issues and generate remediation actions that integrate into release cycles.

Manage your cloud operations

Effective cloud management optimizes operational efficiency, reduces downtime, and clarifies roles and responsibilities. Standardize your cloud operations through automation and structured support processes. Follow these operational guidelines:

- **Provide continuous cloud support.** Establish 24/7 support coverage through either global teams adopting a follow-the-sun model or structured on-call rotations. Define responsibilities clearly to ensure timely response and resolution of critical incidents. Configure automated [alerts](#) to notify designated support personnel immediately.
- **Automate repetitive work.** Use Azure automation capabilities to minimize manual processes and reduce operational overhead. Automate routine activities to eliminate errors, streamline workflows, and enable teams to concentrate on strategic priorities.

[Expand table](#)

Use Case	Examples
Automation	Automate workflows in Azure Boards or ITSM system. Templates for "Change Request" and "Incident" work items.
Incident response	To autogenerate incident tickets with standard fields populated, integrate Azure Monitor and Azure Service Health with ticketing system.
Change management	Use Azure Logic Apps to autoapprove low-risk changes or autoremediate certain incidents.
Compliance	Use Azure Policy to enforce and monitor cloud compliance.
Security	Use Microsoft Defender for Cloud and Microsoft Sentinel to automate security threat detection and response. Use Microsoft Entra ID Governance to review permissions and automate permissions management.

Improve operations

Optimize your Azure cloud environment by promoting continuous improvement. Regularly evaluate operations and prioritize ongoing learning and feedback. Follow these steps:

1. **Review operations to improve.** Follow best practices to [monitor](#) the health, compliance, security, costs, data, and cloud resources. Conduct weekly operational reviews to discuss key metrics, recent incidents, deployed changes, and anticipated risks. Actively address [resource sprawl](#) and [technical debt](#).
2. **Train for operations.** Foster ongoing skill development by prioritizing essential learning resources. Maintain dynamic cloud operations through practical training environments. The

following table provides resources for operations training.

 Expand table

Operations training	Description
Get credentials	Set goals for Microsoft credentials , like applied skills and Microsoft Certifications to build expertise.
Use operational resources	See Azure management resources .
Use product documentation	Use Microsoft Learn to find guidance on Azure services .
Get hands-on practice	Encourage hands-on practice in nonproduction sandbox environments.

Azure management resources

 Expand table

Category	Management resource	Description
Compliance	CAF Govern	Microsoft's cloud governance framework
Security	Manage security operations	Guidance to manage security operations
Security	Microsoft security tool	A list of Microsoft and Azure security tools
Security	Workload security	Workload guidance for security
Resource management	Naming and tagging strategy	Naming and tagging recommendations to manage resources
Resource management	Azure abbreviation	List of abbreviations for Azure resources
Resource management	Azure Advisor	A digital assistant to align with Azure best practices.
Resource management	Azure naming rules	Naming rules for all Azure resources
Resource management	Azure service guides	Guidance for service configuration decisions
Development	Workload software development	Workload guidance for software development
Development	Azure Architecture Center	Architectures and guides for different use cases

Category	Management resource	Description
Development	Developer resource hub	A hub for developer tools and resources
Deployment	Bicep, Terraform, and ARM templates	IaC templates for every Azure resource
Deployment	Azure region pairs	List of Azure paired regions
Deployment	Directory of Azure Cloud Services	Directory of all Azure services
Deployment	Workload deployment	Workload guidance for continuous integration
Monitoring	Monitor your Azure cloud estate	Comprehensive Azure monitoring guidance
Monitoring	Workload monitoring	Workload guidance for monitoring
Cost	Manage costs	Cost management guidance
Cost	Workload cost optimization	Workload guidance for cost optimization
Reliability	Manage data reliability	Guidance to maintain data reliability
Reliability	Manage cloud resource reliability	Guidance to maintain resource reliability
Reliability	Manage security incidents	Recommendations to respond to security incidents
Performance	Workload performance efficiency	Workload guidance for performance efficiency

Next steps

Administer your Azure cloud estate

Last updated on 09/16/2025